

## Objective

**Characterization of the DPD metric with respect to tuning parameter  $\alpha$ .** Given an  $\alpha$  in a minimal Density Power Divergence fitting setup, can I analytically say if the optimal fit will be robust or efficient?

**Details:**

1. I want to fit a candidate distribution  $f(x; \theta)$  over a target density  $g(x; \theta)$  possibly contaminated with outliers, by minimizing the  $DPD_\alpha(f, g)$  divergence over  $\theta \in \Theta$ .
2. Suppose we partition parameter space  $\Theta$  of the candidate density into

$$\Theta = \Theta_r \cup \Theta_e$$

viz. robust and efficient estimates.

3. I intend to find the partitions of  $A = [0, 1] \ni \alpha$ :

$$A = A_r \cup A_e$$

which cause the robust and efficient fits respectively. i.e. upon selecting  $\alpha \in A_r$ , we obtain min. div. estimate  $\theta \in \Theta_r$ .

**I'm hoping this will answer a long-standing question of selection of the tuning parameter for DPD.**

## Progress

I have devised a visualization that I call the “potential map”. See Figure 1

DPD divergence is some integral

$$D(f, g) : \int_{x \in \mathcal{X}} L(f(x), g(x)) dx$$

Call  $V(x, y) = L(y, g(x))$  the **divergence potential associated with  $g$  and induced by  $L$**  (probably bad nomenclature! Some source called it “divergence density”, not sure if that is standard terminology).

Plotting  $V$  (either 3D map or heat-plot) gives a glimpse at the “topography of penalization of  $D(\cdot, \cdot)$  with respect to  $g$ ”, because I think

The process of fitting a candidate distribution over a given target is merely an act of minimizing the line integral of the potential function  $V(x, y)$  along the density line  $f(x; \theta)$ , where the minimization is over  $\Theta$ .

I say line integral but it is not exactly line integral per se.

I have tabulated common divergences with their associated potentials are provided in Table 1. Note how these divergences are functions of  $f - g$ , i.e. dependent on the abscissa-wise deviation of the candidate density from the target. However the **rate of increment of penalization** with increasing abscissa-wise deviation are functionally different across different divergence metrics. Further, for DPD, it is also controllable by its hyperparameter  $\alpha$ .

| Divergence name | Form                                                              | Potential $L_g(y)$                                         | Gradient $\nabla_y L_g(y)$                 |
|-----------------|-------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------|
| Hellinger       | $\int \frac{1}{2}(\sqrt{f(x)} - \sqrt{g(x)})^2 dx$                | $\frac{1}{2}(\sqrt{y} - \sqrt{g})^2$                       | $\frac{1}{2\sqrt{y}}(\sqrt{y} - \sqrt{g})$ |
| DPD- $\alpha$   | $\int f^\alpha(f - g) - \frac{1}{\alpha}g(f^\alpha - g^\alpha)dx$ | $f^\alpha(f - g) - \frac{1}{\alpha}g(f^\alpha - g^\alpha)$ | $(1 + \alpha)y^{\alpha-1}(y - g)$          |

Table 1: Some common PDF-PDF divergences and their potential functions

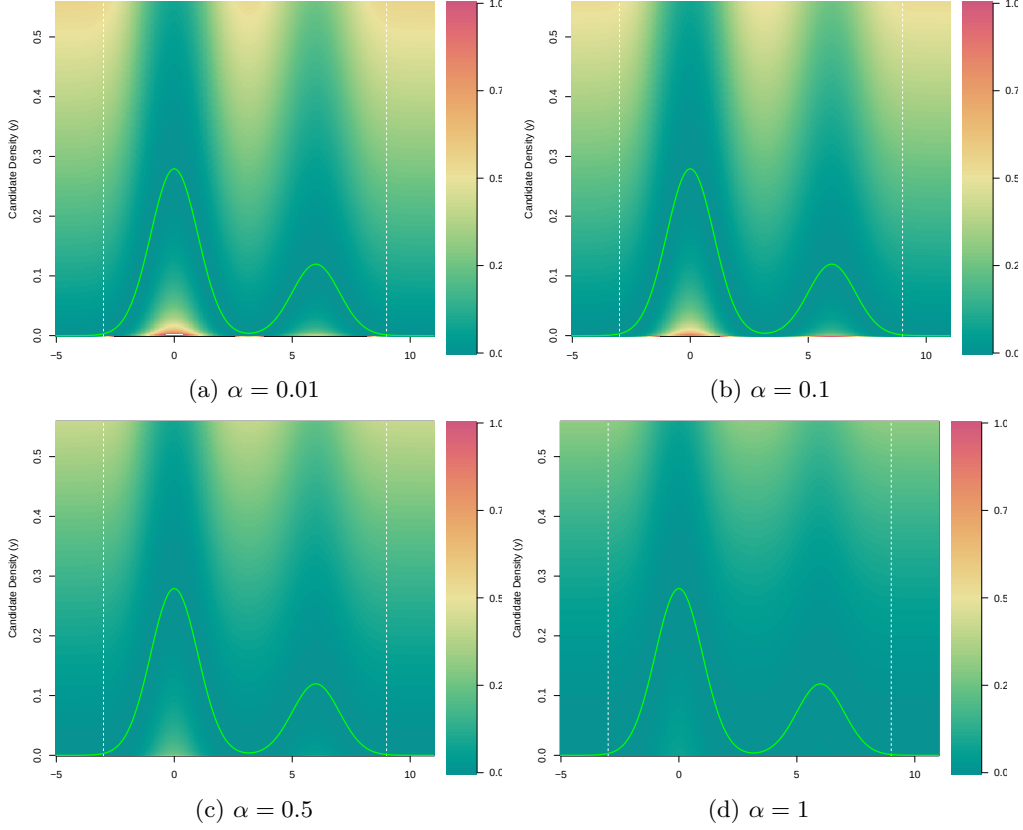


Figure 1: Divergence Potentials created by the same contaminated density for different  $\alpha$ 's of DPD

## A graphical demonstration

### Observation:

If we were to fit (via say numerical optimization) some unimodal  $f$  over this contaminated  $g$ , we would expect to see that:

1. Higher  $\alpha$  induces lower penalization for deviating some amount from the target  $g$ , thereby making an efficient fit that spans over both density peaks more feasible.
2. However, a lower  $\alpha$  allows isolated fitting over one of the 2 peaks only (the principal peak fares better), since otherwise, trying to span both peaks by passing through the middle high-penalization region incurs a lot of penalty/potential.

### Important Takeaway

For contaminated densities, it is not the “attraction” of the second peak that deviates the candidate to fit contaminatedly (i.e. efficiently), but it is the high potential “hill” in between that discourages such a fit.

**Example Formulation:**

Let  $f(x; a, d, s, k) := \frac{1}{dA(s, k)} \sin^k \left( \frac{x-a}{d} \right)^s$  with  $a, d, s, k$  all unknown,  
 $g(x) := p\phi(x) + (1-p)\phi(x-h)$  with  $p, h$  given,  
and for a given  $\alpha$ , the task is to minimize:

$$\min_{s, k} \int_a^{a+d} \left[ f^\alpha(x)(f-g)(x) - \frac{1}{\alpha} g(x)(f^\alpha - g^\alpha)(x) \right] dx$$